

BIOLOGY

Courses

Introductory courses in the Biology Department can serve either as prerequisites for further study for the biology major/minor or satisfy the Nature Sciences area course requirement for General Education. The department offers one introductory course intended for non-majors.

Non-introductory courses are designed for students seeking more in-depth information on cellular and molecular, organismal, and population biology and for students with specific vocational aims.

See the [course catalog](#) for descriptions and updated information.

The Biology Major culminates in the senior capstone experience, where students pursue their own investigation into a biological question of their choosing. For most students, this capstone experience will comprise the 4-credit capstone course, taken in the student's senior year.

Alternatively, students may conduct faculty-mentored research for credit (≥ 3 credits total), then enroll in the 1-credit capstone course in their senior year. All students who are enrolled in the 1-credit capstone course must complete the [1-hour Capstone Contract form](#) by the end of the first week of that course. Students who would like to count research mentored by faculty in another department (ex: chemistry, kinesiology, psychology) towards their capstone experience in biology must complete the [Interdisciplinary Capstone Request for Biology form](#) and receive approval. (Ideally, this form should be submitted as soon as possible after initiating your faculty-mentored research experience.)

50-114 Explorations in Biology

The course approach will combine inquiry-based learning with content delivery and an integration of laboratory skills. Explorations in Biology provides a natural science course

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topics in the field of biology. Together, this course helps promote observation, experimentation and analytical skills. This course is intended for students seeking to fulfill general education requirements. Contributes to Health Studies. (NS)

50-154 Climate Change

This course examines the science behind climate change, current and future impacts on the natural world, and current strategies to mitigate climate change. Students will discuss the impacts of climate change on the cryosphere, extreme weather events, food security and biodiversity. Students will compare and contrast different mitigation and adaptation strategies and discuss the efficacy of each. Contributes to Environmental Studies. (NS)

50-171 Inquiries in Biology

This course provides students an opportunity to participate in the process of science by developing and applying laboratory skills to questions in biology. Contributes to Animal Studies, Health Studies, and Spanish for Healthcare Professions Certificate. To be taken concurrently with either Biology 50-173 or 50-183. (NSL)

50-173 Molecular & Cellular Foundations of Biol

This course introduces students to fundamental concepts in molecular and cell biology, including the molecular basis of genetic information, gene expression, cell structure and function, and cellular bioenergetics. This course is required for students majoring in Biology or seeking a Bachelor's of Science. Not intended for students solely seeking to fulfill general education requirements. Contributes to Animal Studies, Health Studies, and Spanish for Healthcare Professions Certificate. To be taken concurrently with Biology 50-171 or 50-181. (NS)

50-181 Investigations in Biology

This course provides students with an opportunity to participate in the process of science by applying knowledge and concepts to investigation of biological processes. Contributes to Animal Studies, Environmental Studies, Health Studies, and Spanish for Healthcare Professions Certificate. To be taken concurrently with either Biology 50-173 or 50-183.

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This course introduces students to the diversity of life, how different organisms interact with the environment and with each other, principles of heredity, and microevolutionary processes. This course is required for students majoring in Biology or seeking a Bachelor's of Science. Not intended for students solely seeking to fulfill general education requirements. Contributes to Animal Studies, Environmental Studies, Health Studies, and Spanish for Healthcare Professions Certificate. To be taken concurrently with Biology 50-171 or 50-181. Prerequisite: Biology 50-173. (NS)

50-194 Science of Chocolate

This course explores the biology and chemistry of *Theobroma cacao*, the plant from which people make chocolate. Topics in the course will include pollination ecology, genetics, antioxidants and health and climate change. In addition, production versus consumption of chocolate will be discussed in light of economics and social justice issues including child slavery, indigenous communities and agricultural goods. Class activities will include a combination of inquiry-based projects, class discussion and quantitative reasoning. Contributes to Environmental Studies. (SJ) (NS)(PJ)

50-222 Methods in Ecology and Evolutionary Biology (3-3; Half Semester)

This lecture/laboratory course is a foundation-building course that contains instruction on reading the primary literature in ecology and evolutionary biology, conducting literature searches, designing experiments, writing scientific papers, using quantitative methods, exercising critical thinking skills for data analyses, creating graphs, and developing specific laboratory and field research skills for ecology and evolutionary biology. Contributes to Data Analytics and Data Science. Prerequisite: Biology 50-173/171 and 50-183/181, and Mathematics 52-114. (Fall and Spring) (NSL) (WA)

50-232 Methods in Cellular/Molecular Biology (3-3; Half-Semester)

This lecture/laboratory course is a foundation-building course that contains instruction on reading the primary literature in cellular/molecular biology, conducting literature searches, designing experiments, writing scientific papers, using quantitative methods, exercising

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Prerequisites: Biology 50-175/171, 50-185/181, and Chemistry 51-100. Chemistry 51-203 is recommended. (Fall and Spring) (NSL) (WA)

50-251 Integrative Approach to Human A&P I

See Kinesiology 48-251. This course is required for all students pursuing a BS in Kinesiology but may be taken by students seeking to fulfill general education requirements. Also Kinesiology 48-251. Contributes to Health Studies. This laboratory course must be taken concurrently with Biology 50-253. Prerequisites: Kinesiology 48-123/121. (NS)

50-253 Human Anatomy & Physiology I

This course is required for all students pursuing a BS in Kinesiology but may be taken by students seeking to fulfill general education requirements. Also Kinesiology 48-253. Contributes to Health Studies. This course must be taken concurrently with Biology 50-251. Prerequisites: Kinesiology 48-123/121. (NS)

50-261 Integrative Approach to Human A & P II

See Kinesiology 48-261. This course is required for students pursuing a BS degree in Kinesiology. It is not intended for students solely seeking to fulfill general education requirements. This is the second of a two-course sequence in the study of human anatomy and physiology. Also Kinesiology 48-261. Contributes to Health Studies and Neuroscience. This laboratory course must be taken concurrently with Biology 50-263. Prerequisites: Kinesiology 48-253/251 or Biology 50-253/251. (NS)

50-263 Human Anatomy & Physiology II

See Kinesiology 48-263. This course is required for students pursuing a BS degree in Kinesiology. It is not intended for students solely seeking to fulfill general education requirements. This is the second of a two-course sequence in the study of human anatomy and physiology. Contributes to Health Studies and Neuroscience. Also Kinesiology 48-263. Prerequisites: Kinesiology 48-253/251 or Biology 50-253/251. (NS).

50-314 Genetics

An introduction to the study of genetics, including the principles of heredity, structure and

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population genetics. The course includes discussion of current findings of genetic research. Laboratory exercises emphasize hypothesis testing and the analysis of genetic crosses, along with techniques and concepts of genetics. Contributes to Health Studies. Prerequisites: Biology 50-173/171 and 50-183/181; and one of the following: Biology 50-222, Biology 50-232, Kinesiology 48-214, Psychology 33-204, 33-214, or 33-224. (Cellular and Molecular Biology, Ecology and Evolutionary Biology) (NSL)

50-324 Behavioral Neuroscience

See Psychology 33-534. Contributes to Animal Studies, Health Studies, and Neuroscience. Prerequisite: Take Biology 50-173/171 or 50-183/181, and one of the following: Psychology 33-204, Psychology 33-214, Psychology 33-224, Kinesiology 48-214, Biology 50-222 & 50-232, or Chemistry 51-862; or permission of instructor. (Cellular and Molecular Biology)(Spring)

50-334 Evolutionary Biology

An exploration of the possible mechanisms of evolution. Topics to be discussed include natural selection, punctuated evolution, population genetics, adaptation, units of selection, speciation, evolutionary biogeography and macroevolution. Contributes to Animal Studies. Prerequisite: Biology 50-173/171 and 50-183/181, and one of the following: Biology 50-222, Kinesiology 48-214, or Psychology 33-204, 33-214, or 33-224. (Ecology and Evolutionary Biology) (NS)

50-354 Neurobiology

A deep dive into the biology of the nervous system, focusing on cellular, molecular, and systems neuroscience. Students will learn the fundamentals of neurobiology, with an emphasis on the relationship between structure and function in the nervous system. Students will also discuss research at the cutting-edge of neuroscience, as well as the interface between neurobiology, ethics, and society. Contributes to Neuroscience. Prerequisite: Biology 50-173/171 and 50-183/181, and one of the following: Biology 50-232, Chemistry 51-862, Kinesiology 48-214, or Psychology 33-204, 33-214, or

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After a brief consideration of the lower chordates, this course deals with the functional anatomy of the vertebrates. Although there is some study of vertebrates in natural environments, primary emphasis is on laboratory dissections of preserved specimens. Contributes to Animal Studies and Computer-Aided Design & 3D Printing Certificate. Prerequisite: Biology 50-173/171 and 50-183/181, and one of the following: Biology 50-222, Biology 50-232, Kinesiology 48-214, or Psychology 33-204, 33-214, or 33-224. (NSL)

50-374 Human Genetics and Evolution

An exploration of major themes and case studies in human variation and human evolution, including polygenic traits, gene by environment interaction, molecular evidence of natural selection, epigenetics, and variation in life history traits. Prerequisite: Biology 50-173/171 and 50-183/181, and one of the following: Biology 50-222, Kinesiology 48-214, or Psychology 33-204, 33-214, or 33-224. (Ecology and Evolutionary Biology) (NS)

50-384 Conservation Biology

A survey of the field of conservation biology through global and local conservation case studies, including threats to biodiversity, solutions to conservation problems and protection of endangered species. Contributes to Animal Studies and Environmental Studies. Prerequisite: Biology 50-173/171 and 50-183/181, and Biology 50-222, Kinesiology 48-214, or Psychology 33-204, 33-214, or 33-224. (Ecology and Evolutionary Biology) (NS)

50-394 Developmental Biology

An exploration of the principles and processes that generate a complex organism from a single cell. Students will take a guided tour of early animal development and organogenesis, emphasizing the self-organizing nature of the embryo. Students will also draw connections between developmental biology and evolution. Weekly lab sessions will include a combination of benchwork with living organisms and group discussions of related topics. Contributes to Neuroscience. Prerequisite: Biology 50-173/171 and 50-183/181, and one

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50-404 Cancer Biology (3-0)

An interdisciplinary approach to the study of human cancer, including in-depth analysis of the variety of molecular mechanisms that contribute to cancer development and progression, examination of how biomedical research findings are translated into clinical practice and issues related to patient care. Contributes to Health Studies. Prerequisite: Biology 50-173/171 and 50-183/181, and one of the following: Biology 50-232 or Chemistry 51-862. (Cellular and Molecular Biology) (NS)

50-414 Plant Ecology

This course explores the diversity within the plant kingdom using field and laboratory experiences. Students will be introduced to the structure and function of plants through an investigation of plant cells, tissues, organs, and basic physiological processes. Students will also discuss the economic importance, human uses, and significance of plants to society. Contributes to Environmental Studies. Prerequisite: Biology 50-173/171 and 50-183/181, and one of the following: Biology 50-222, Kinesiology 48-214, Psychology 33-204, 33-214, or 33-224, or Environmental Studies 49-224, or instructor permission. (Ecology and Evolutionary Biology) (NSL)

50-424 Organ Physiology (3-3)

Processes/functions of organ systems: nervous, muscular, cardiac, circulatory, respiratory, renal, digestive and endocrine. Human physiology is emphasized. Contributes to Health Studies and Neuroscience. Prerequisite: Biology 50-173/171 and 50-183/181, and one of the following: Biology 50-232, Chemistry 51-862, Kinesiology 48-214, or Psychology 33-204, or permission of instructor. (Cellular and Molecular Biology) (NSL)

50-434 Ecology (3-3)

This class explores the interactions of organisms with their biotic and abiotic environment. In particular, the course looks at the influence of nutrients, climate, competition, predation and symbiotic relationships on individuals, populations and communities. Contributes to Animal Studies and Environmental Studies. Prerequisite: Biology 50-173/171 and

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50-444 Invertebrate Ecology (5-5)

This class explores the amazing diversity found across marine, terrestrial and aquatic habitats. The lecture component involves taxonomic descriptors of different groups, but more specifically focuses on the ecology of these organisms through critical reading of the primary literature. Through the semester, the course confronts topics that impact many invertebrates, such as exotic species, habitat degradation, chemical communication, predator-prey interactions and competition. In weekly lab sessions, special emphasis is placed on conducting experiments, learning to identify organisms, and investigating the role of aquatic insects in ponds and streams through field work. Contributes to Animal Studies. Prerequisite: Biology 50-173/171 and 50-183/181, and one of the following: Biology 50-222, Kinesiology 48-214, or Psychology 33-204, 33-214, or 33-224. Biology 50-434 is recommended but not required. (Ecology and Evolutionary Biology) (NSL)

50-454 Tissue Mechanics

See Kinesiology 48-454. Contributes to Health Studies. (NS)

50-464 Molecular Biology

This course focuses on the molecular aspects of genetic systems in prokaryotes, eukaryotes and viruses. Topics include: molecular methods and their applications, cell cycle control, gene expression, regulation of gene expression, gene arrangement, DNA mutagenesis and repair, mobile genetic elements and viral replication. Laboratory includes independent projects using molecular biology techniques. Prerequisite: Biology 50-173/171 and 50-183/181, Chemistry 51-313, and one of the following: Biology 50-232 or Chemistry 51-862. (Cellular and Molecular Biology) (NSL)

50-474 Genetics, Genomics, and Medicine

An exploration of human phenotype, including Mendelian, polygenic and environmental influences, using approaches ranging from family studies and evolutionary medicine to population genomics. The lab will focus on bioinformatic resources in genetics, genomics, and medicine. Contributes to Data Analytics and Data Science. Prerequisite: Biology

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(NSL)

50-484 Microbiology (3-3)

An introduction to the study of microbes. This course is not strictly a bacteriology course as some attention is given to fungi and viruses. This course includes microbial cell structure and function, growth, metabolism and genetics. Microbial diversity is a recurring theme throughout the course. The course includes a weekly laboratory session. Prerequisite: Biology 50-173/171 and 50-183/181, Chemistry 51-313, and one of the following: Biology 50-232 or Chemistry 51-862. or permission of instructor. (Cellular and Molecular Biology) (NSL)

50-494 Biology of Reproduction

This course takes a comprehensive look at the process of reproduction by examining the role of hormones, developmental and genetic sex, the process of puberty, and the production of offspring. Emphasis is given to human reproduction, although other species are studied to assist in the understanding of reproduction. Contributes to Neuroscience. Biology 50-173/171 and 50-183/181, and one of the following: Biology 50-232 or Chemistry 51-862. (Cellular and Molecular Biology) (NS)

50-514 Cellular Physiology (3-3)

The general functions of eukaryotic cells are studied primarily in animal cells. Topics include transcription, translation, protein functions, cell motility, secretion and endocytosis, cell signaling and cell cycling. Laboratory experiments teach techniques and concepts of cellular physiology. Contributes to Health Studies. Prerequisite: Biology 50-173/171 and 50-183/181, Chemistry 51-313, and one of the following: Biology 50-232 or Chemistry 51-862. (Cellular and Molecular Biology) (NSL)

50-524 Endocrinology (3-3)

This course undertakes a detailed exposure to the structure and function of the endocrine system. The course emphasizes the biosynthesis, mechanism of action and homeostatic function of hormones. Topics demonstrate the chemical and physiological principles of

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50-232 or Chemistry 51-862, or permission of the instructor. (Cellular and Molecular Biology) (NSL)

50-534 Fundamentals of Immunology (3-0)

An introduction to the immune system as studied in mammals. Emphasis is placed on acquired immunity, specifically as it pertains to the humoral and cell-mediated immune responses. This course deals with the cellular and biochemical mechanisms involved in the education and regulation of both the humoral and cell-mediated immune responses.

Prerequisite: Biology 50-173/171 and 50-183/181, Chemistry 51-313, and one of the following: Biology 50-232 or Chemistry 51-862. Chemistry 51-334 and 51-604 are recommended but not required; or permission of instructor. (Cellular and Molecular Biology) (NS)

50-604 Principles of Biochemistry

Also Chemistry 51-604. This course provides an overview of the major classes of biological macromolecules (proteins, carbohydrates, lipids and nucleic acids) and their functions in cellular structure, information pathways, and enzyme catalysis, and bioenergetics. This foundation will support an introduction to metabolism and key concepts in regulation of central metabolic pathways. This course is designed for students majoring in one of the natural sciences but who do not require a two-semester course in biochemistry. This course should not be taken by chemistry or biochemistry majors. Approved as a cellular/molecular course for use in the Biology major. Prerequisite: Chemistry 51-323. (NS) (Fall)

50-624 General Biochemistry II

Also Chemistry 51-624. This course focuses on the metabolism (break-down and formation) of carbohydrates, lipids and proteins in the human body and the energy involved with such processes. It is the second course of a two-semester comprehensive sequence and is designed specifically for chemistry and biochemistry majors as well as those students interested in pursuing scientific research careers. Approved as a cellular/molecular course for use in the Biology major. Prerequisites: Chemistry 51-604 or Chemistry 51-614. (NS)

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Students must plan with a faculty member in the Biology Department prior to enrolling in this course. May contribute to Neuroscience. Prerequisite: Permission of the instructor and a completed course description report. May be repeated with changed content.

50-972 Research in Biology.

Students must plan with a faculty member in the Biology Department prior to enrolling in this course. May contribute to Neuroscience. Prerequisite: Permission of the instructor and a completed course description report. May be repeated with changed content.

50-973 Research in Biology

Students must plan with a faculty member in the Biology Department prior to enrolling in this course. May contribute to Neuroscience. Prerequisite: Permission of the instructor and a completed course description report. May be repeated with changed content.

50-974 Research in Biology

Students must plan with a faculty member in the Biology Department prior to enrolling in this course. May contribute to Neuroscience. Prerequisite: Permission of the instructor and a completed course description report. May be repeated with changed content.

50-991 Biology Capstone Seminar

Topics vary by interest, but biology students will expand their knowledge of the scientific literature, write a research paper and present their work to their peers. Biology majors that have pursued at least three hours of total credit in Research in Biology (BIO50-97X) will have the opportunity to complete their research paper and present their work in an integrated one credit hour version of the seminar. Biology majors that have not pursued three hours of total credit in Research in Biology (BIO50-97X) will take the four-hour version of the seminar. Accordingly, the Biology Capstone may involve group and/or individual research projects. The seminar will also provide time and preparation for all students to take the Biology Major Field Test (MFT). (Fall and Spring). Pre-requisite: Senior Standing.

50-994 Biology Capstone Seminar

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have pursued at least three hours of total credit in Research in Biology (BIO50-97X) will have the opportunity to complete their research paper and present their work in an integrated one credit hour version of the seminar. Biology majors that have not pursued three hours of total credit in Research in Biology (BIO50-97X) will take the four-hour version of the seminar. Accordingly, the Biology Capstone may involve group and/or individual research projects. The seminar will also provide time and preparation for all students to take the Biology Major Field Test (MFT). (Fall and Spring). Pre-requisite: Senior Standing.

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CONTACT